

Who leaves teaching?

Teacher attrition in the United States, 2005–2025

EXECUTIVE SUMMARY

- Each year, **15.1%** of degree-holding school teachers leave the profession and do not return within our observation window. The conventional 12-month measure gives 17.6%, but one in six of those leavers is teaching again within three months.
- Persistent attrition rose from about 13% in 2005–2010 to about 17% in 2022–2024, with a peak in the pandemic year and a new high of 18% in 2024. The rise is almost entirely a rise in switching to other occupations.
- Part-time work (+11 percentage points) and private-sector employment (+7) are the strongest predictors of leaving, in every subsample, and higher pay retains. A new baby during the year does not move men's exit probability but raises women's by 5 points, and most new mothers who leave exit the labor force entirely.
- The typical leaver is a woman in her late career, working part-time, more often in a private school. The highest-risk decile has a 37% exit probability, 2.5 times the average.

15.1 %

of teachers leave the profession for good each year

+11 pp

extra exit risk from part-time work

1 in 6

12-month leavers is back teaching within 3 months

1. Data

Our data are the basic monthly microdata of the Current Population Survey (CPS), the household survey behind the official employment statistics of the United States, which we assemble for every month between January 2005 and December 2025.¹ What makes the CPS useful for studying attrition is its rotation design, illustrated in Figure 1. Each household is interviewed during four consecutive months, rests for eight, and is interviewed again during four more, so that any person can be observed for at most sixteen months and only once in a lifetime. Within that window, an adult interviewed in year t is re-interviewed in the same calendar month of $t+1$, and the two interviews can be linked through household and person identifiers, with every match validated on sex, race and a plausible age progression. Because nobody is followed across several years, each base year forms a fresh cohort, and every rate in this brief is computed wave by wave over twenty waves.

¹The files come from the U.S. Census Bureau for 2024–2025 and from the NBER mirror for 2005–2023. October 2025 was never released because the federal government shutdown suspended that month's data collection.

Figure 1: The CPS follows each person for at most 16 months.



Notes: Interview calendar of one rotation group; numbers are months since the first interview. The 12-month link compares an interview from the first block with the interview held exactly one year later, and the return window uses the remaining interviews of the second block.

Table 1 traces how this design narrows the CPS universe down to our analytic sample. In an average month the survey interviews close to 95,000 adults who represent 241 million people, of whom about 1,850 are employed school teachers with at least a bachelor’s degree, representing the 4.6 million teachers of the country. Teachers are identified by the occupation of their main job, which covers preschool and kindergarten, elementary and middle school, secondary school and special education (occupation codes 2300 to 2330, identical across the three classification vintages the period spans), and the degree requirement, standard in the attrition literature, removes a small group of daycare and substitute personnel. Only the half of the sample that is in its first rotation block can be linked one year ahead. The linkage itself succeeds for 78 percent of teachers, and reassuringly it succeeds less often for everyone else, 74 percent among other college-educated workers and 71 percent among all employed, a ranking that holds in every single year. Losses arise because the CPS samples addresses rather than people, so movers drop out, but they do not select against teachers, and because linked persons keep their population weights, all weighted rates remain representative of the 4.6 million teachers. Pooling the twenty waves delivers 174,872 linked teachers, of whom 129,897 have at least one further interview after the 12-month link and form our main sample.

Table 1: From the CPS universe to the analytic sample.

	Persons	Population represented
Adults interviewed in an average month	94,630	240.6M
employed	57,572	148.1M
school teachers (occ. 2300–2330)	2,094	5.2M
with bachelor’s degree or higher	1,849	4.6M
Linked 12 months later (unique persons, 20 waves)	174,872	the same 4.6M
with re-interviews after $t+12$: main sample	129,897	the same 4.6M

Notes: Monthly figures are averages over the 251 available months, with weighted population totals beside the sample counts. Teachers are employed persons whose main job is school teaching (Census occupation codes 2300 to 2330) holding a bachelor’s degree or higher. Source: authors’ calculations from CPS basic monthly microdata, 2005–2025.

A leaver could simply be anyone not employed as a school teacher twelve months after baseline, which is the convention of the NCES Teacher Follow-up Survey, but temporary exits such as parental leave or a short unemployment spell inflate that measure, and indeed 15.7% of such leavers are teaching again within three months. Our main outcome is therefore the persistent leaver, a teacher who is out of teaching at $t+12$ and in every later re-interview, computed on the 129,897 teachers whose rotation allows at least one re-interview. Exits longer than a year that end

in a return remain undetectable, so persistent attrition is still an upper bound on permanent exit, only a much tighter one. A second caveat concerns measurement. Occupation in the returning rotation is re-coded independently, which inflates switching, so levels should be read as upper bounds while comparisons across groups and over time, the object of this brief, are far less affected.

The richness of the linked file is summarized in Table 2, which compares stayers and persistent leavers across every characteristic we observe, with the differences that are both statistically significant and economically meaningful shaded in green. Read as a whole, the table already sketches the profile of the leaver. Leavers are three years older on average, they are far more attached to the margins of the job than to its core, since working part-time is eleven points more common among them and private-sector employment fifteen points more common, and their family situation is different in a precise way, with fewer young children at home rather than more. The gender split, in contrast, is economically irrelevant, with women 76.8% of stayers and 77.0% of leavers, an early sign that gender operates through specific channels rather than through levels. Black and non-citizen teachers are clearly over-represented among leavers, and so are preschool and kindergarten teachers, while credentials matter in both directions, since leavers are less likely to hold a master's degree but almost twice as likely to hold a professional degree or doctorate. Most of these gaps are not marginal, as ten of them exceed two percentage points, the threshold we use for economic relevance.

Table 2: Descriptive statistics of the analytic sample (weighted means).

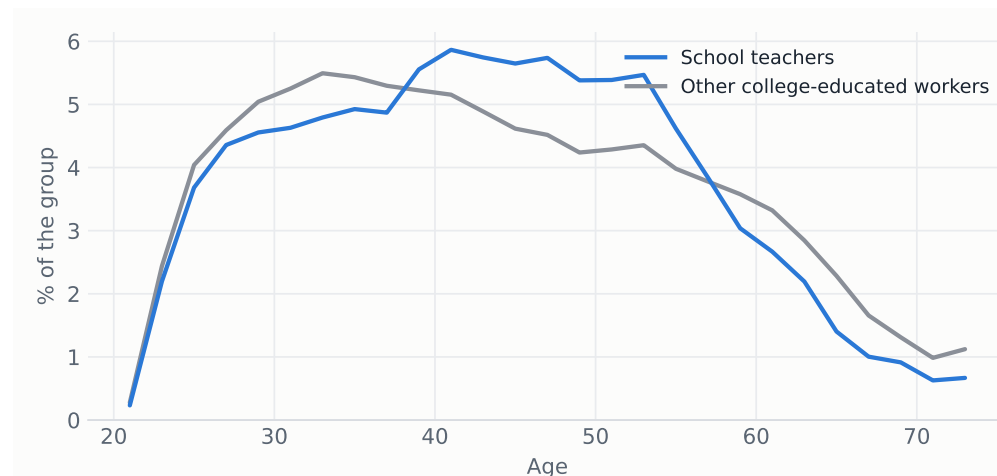
	All teachers	Stayers	Persistent leavers	Difference
Age, years	44.0	43.5	46.7	+3.15***
Female	76.8%	76.8%	77.0%	+0.2 pp
Female aged 25–44	37.8%	38.8%	31.7%	-7.1 pp***
Married	72.0%	72.5%	69.5%	-3.0 pp***
Number of own children (<18) at home	0.8	0.9	0.6	-0.22***
Child under 6 at home	18.1%	18.9%	13.9%	-5.0 pp***
New baby during the year	2.5%	2.5%	2.8%	+0.3 pp
Black	8.1%	7.5%	11.6%	+4.2 pp***
Hispanic	7.9%	7.8%	8.5%	+0.7 pp*
Non-citizen	1.7%	1.5%	3.0%	+1.5 pp***
Master's degree or higher	52.9%	53.7%	48.4%	-5.3 pp***
Professional degree or doctorate	2.8%	2.5%	4.4%	+2.0 pp***
Part-time (<35 h/week)	9.0%	7.2%	19.1%	+11.9 pp***
Holds more than one job	8.1%	7.9%	9.4%	+1.5 pp***
Public-sector employer	78.3%	80.5%	65.7%	-14.9 pp***
Family income \$75k+	76.4%	77.3%	71.4%	-5.9 pp***
Elementary / middle school	61.8%	61.9%	61.2%	-0.7 pp
Preschool / kindergarten	7.7%	7.1%	10.6%	+3.5 pp***
Secondary school	23.1%	23.5%	20.5%	-3.0 pp***
Special education	7.5%	7.5%	7.6%	+0.1 pp
Persons	129,897	110,720	19,177	

Notes: The sample is all linked teachers whose rotation includes at least one interview after the twelve-month link. All variables are measured at baseline and weighted with CPS person weights. Children are own children under eighteen living in the household at the time of the interview. The last column reports the stayer-leaver difference and the significance of a two-sample test with standard errors clustered by household (** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$); green cells mark differences that are significant at 5 percent and larger than two percentage points (one year of age, 0.15 children). Source: authors' calculations from CPS basic monthly microdata, 2005–2025.

2. The teacher workforce

Before asking who leaves, it helps to know who teaches, and the comparison that makes teachers legible is the rest of the college-educated workforce. Demographically, teaching is a distinct segment of the graduate labor market. As Figure 2 shows, its age distribution piles up in mid-career, with a thicker mass between 35 and 55 and thinner tails at both ends than among other graduates, while the balance in Table 3 adds that teachers are overwhelmingly women, at 76 against 49 percent, disproportionately white, and markedly less likely to be Asian or foreign-born. They are more often married and have more children at home.

Figure 2: Age distribution of school teachers and of other college-educated workers, 2021–2025.



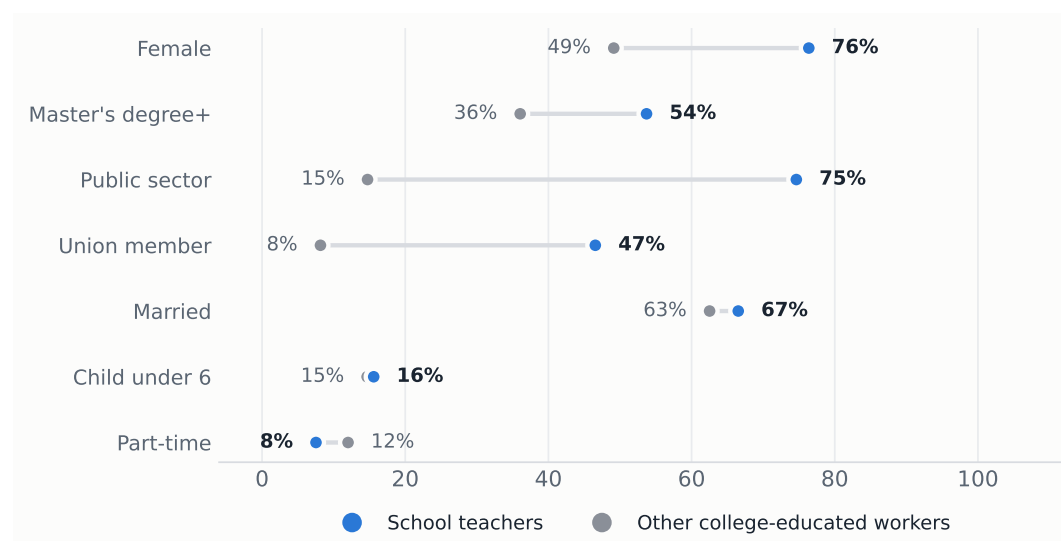
Notes: Both groups are employed persons with at least a bachelor's degree, and teachers are identified as in Table 1. Weighted to population. Source: authors' calculations from CPS basic monthly microdata, 2021–2025.

As a job, teaching is even more distinctive. Figure 3 condenses the contrast. Three quarters of teachers work for the public sector against 15 percent of comparable graduates, 47 percent are union members against 8 percent, and half hold a master's degree, twice the rate of other graduates. Hours are standard full-time, part-time work is less common than elsewhere, and the price of all this stability is pay, since the median teacher earns \$1,200 a week, 20 percent less than the \$1,500 of comparable college-educated workers, a gap the full balance in Table 3 shows to be highly significant.

Weighted to population, the stock is remarkably stable at about 4.7 million teachers over the whole period (Figure 4), with the composition by teaching level equally steady and dominated by elementary and middle school (shown in Appendix Figure A1). Stability of the stock, however, hides a great deal of movement underneath. Tracking the linked pairs in both directions, Figure 5 shows that teaching loses and recruits between 15 and 20 percent of its stock every single year, so the profession is best pictured as a large revolving door whose two flows roughly balance, although in recent years exits have run persistently above entries.

Geography adds a final layer to the portrait. As the map in Figure 6 shows, teachers are a visibly larger share of state employment in New England, New York and parts of the rural South and Plains, up to 3.8 percent in Connecticut, Vermont or Mississippi, and a smaller one in Nevada, Florida and much of the West Coast, down to 2.2 percent. This variation turns out to be related to what schools produce. Linking each state to its official NAEP grade-8 mathematics score, the standard state measure of school quality, Figure 7 shows that states with better scores retain

Figure 3: Composition of the teaching workforce vs other college-educated workers, 2021–2025.



Notes: Same groups as Figure 2. Union membership is only asked in the outgoing rotations. Shares are weighted to population. Source: authors' calculations from CPS basic monthly microdata, 2021–2025.

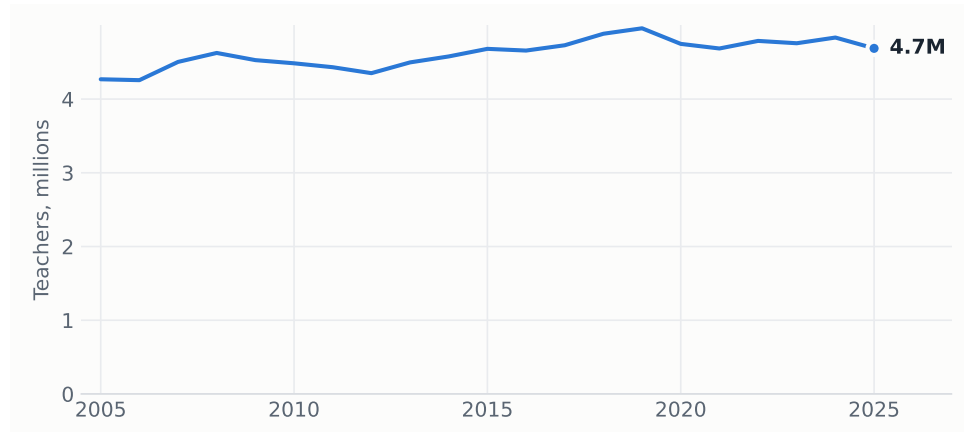
Table 3: Balance: school teachers vs other college-educated workers, 2021–2025 (weighted).

	School teachers	Other college-educated workers	Difference
Age, years	43.5	43.9	-0.44***
Female	76.4%	49.1%	+27.2 pp***
White	83.8%	75.4%	+8.5 pp***
Black	9.8%	10.2%	-0.4 pp
Asian	3.9%	11.7%	-7.7 pp***
Hispanic	10.5%	10.6%	-0.1 pp
Non-citizen	2.5%	7.7%	-5.2 pp***
Married	66.5%	62.5%	+4.0 pp***
Number of own children (<18) at home	0.8	0.6	+0.19***
Child under 6 at home	15.6%	14.7%	+0.9 pp***
Master's degree	50.4%	25.8%	+24.6 pp***
Professional degree or doctorate	3.3%	10.3%	-7.0 pp***
Public-sector employer	74.6%	14.7%	+59.9 pp***
Union member ^a	46.6%	8.2%	+38.4 pp***
Usual weekly hours	40.2	39.8	+0.40***
Part-time (<35 h/week)	7.5%	12.0%	-4.5 pp***
Holds more than one job	7.3%	6.2%	+1.2 pp***
Weekly earnings, median ^a	\$1,200	\$1,500	-\$300***
Family income \$75k+	85.3%	85.5%	-0.2 pp
Persons (monthly interviews pooled)	85,952	1,059,720	

Notes: Both groups are employed persons with at least a bachelor's degree. The last column reports the teacher minus non-teacher difference and the significance of a two-sample test with standard errors clustered by household (** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$). ^a Union membership and weekly earnings are only asked in the outgoing rotations, months in sample 4 and 8. Source: authors' calculations from CPS basic monthly microdata, 2021–2025.

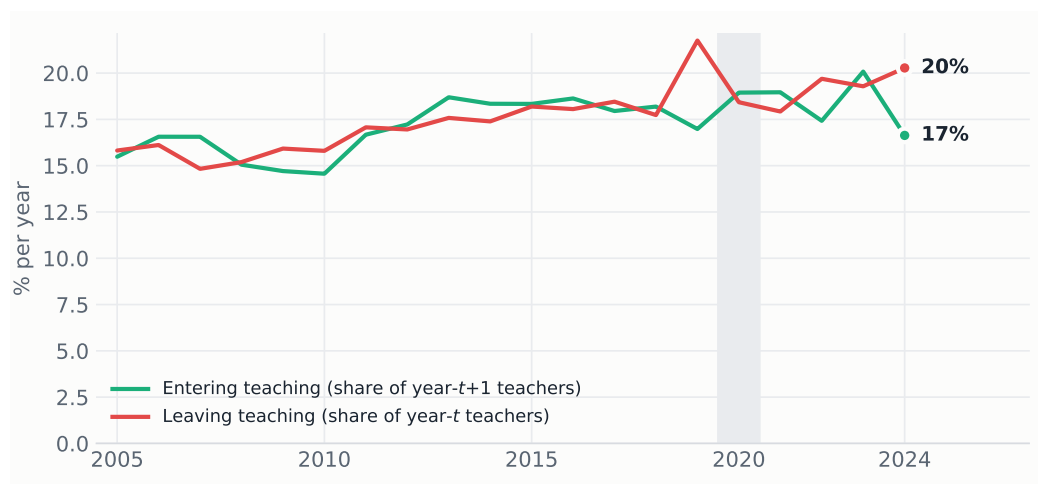
their teachers better, with a correlation of -0.47 and roughly one percentage point less attrition for every ten NAEP points. The direction of causality is of course open, since good schools may retain teachers and stable teaching forces may also produce good schools, but the association places teacher retention squarely inside the education-quality conversation.

Figure 4: The stock of school teachers.



Notes: Employed school teachers with a bachelor's degree or higher, annual average of monthly weighted estimates. Source: authors' calculations from CPS basic monthly microdata, 2005–2025.

Figure 5: Teachers entering and leaving each year.



Notes: Gross annual flows in the linked pairs under the 12-month definitions. A person enters teaching if she is a degree-holding school teacher at the second interview but was not one a year earlier, and leaves in the reverse case. Because the linked sample misses people who move, both flows are conservative. Source: authors' calculations from linked CPS pairs, 2005–2025.

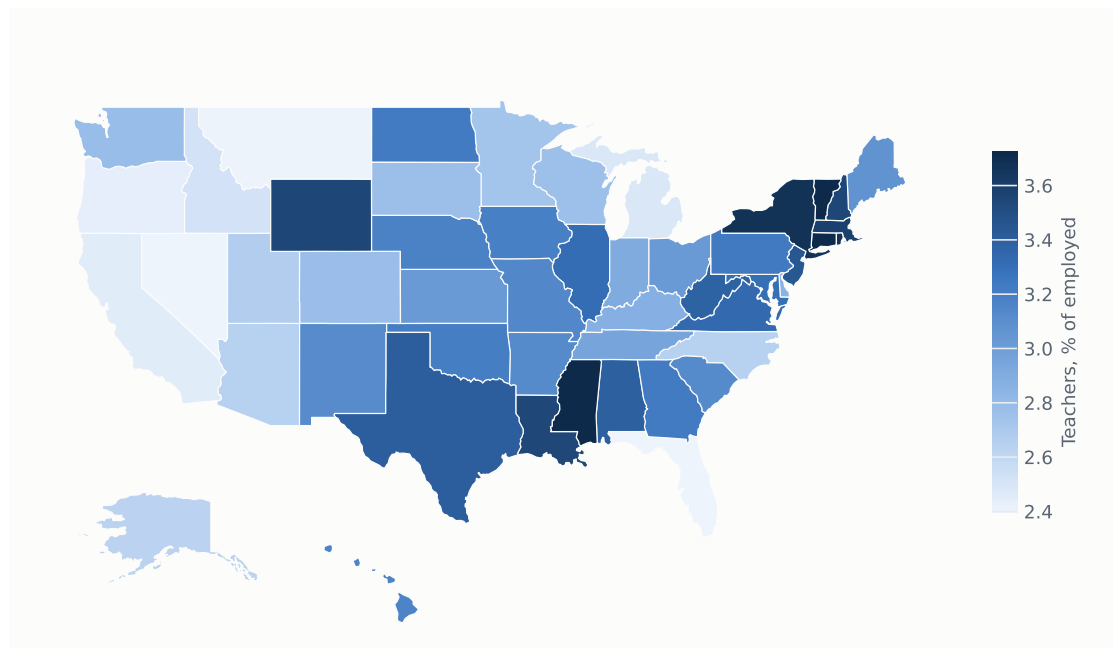
3. Attrition over time

One in six 12-month leavers is teaching again within three months, 16.1 percent of women and 14.0 percent of men, and these temporary exits are precisely what the persistent definition strips out. Net of that churn, Figure 8 shows that persistent attrition still climbed from about 13 percent in 2005–2010 to about 17 percent in 2022–2024, peaking at 18.4 percent in the pandemic year and reaching a new high of 18.0 percent in 2024. Gender gaps are small and unstable throughout, and in the most recent year men, at 19.8 percent, actually exceeded women, at 17.4.

4. Where leavers go

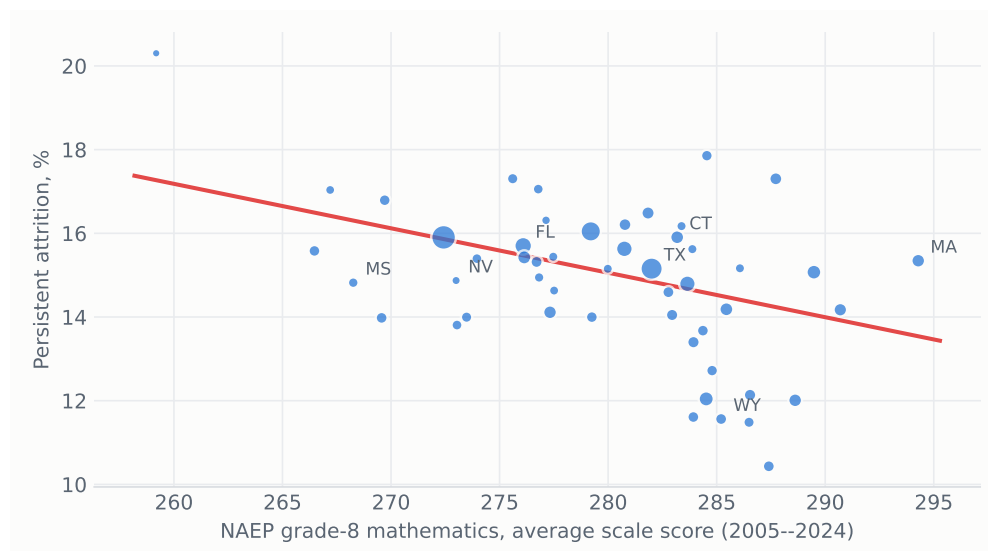
The linked design distinguishes those who abandon the labor force from those who change occupation, and for the latter it records the destination. Figure 9 shows that the single largest

Figure 6: Teachers as a share of state employment, 2021–2025.



Notes: Employed school teachers with a bachelor's degree or higher as a share of total state employment, weighted and pooled over 2021–2025. Source: authors' calculations from CPS basic monthly microdata.

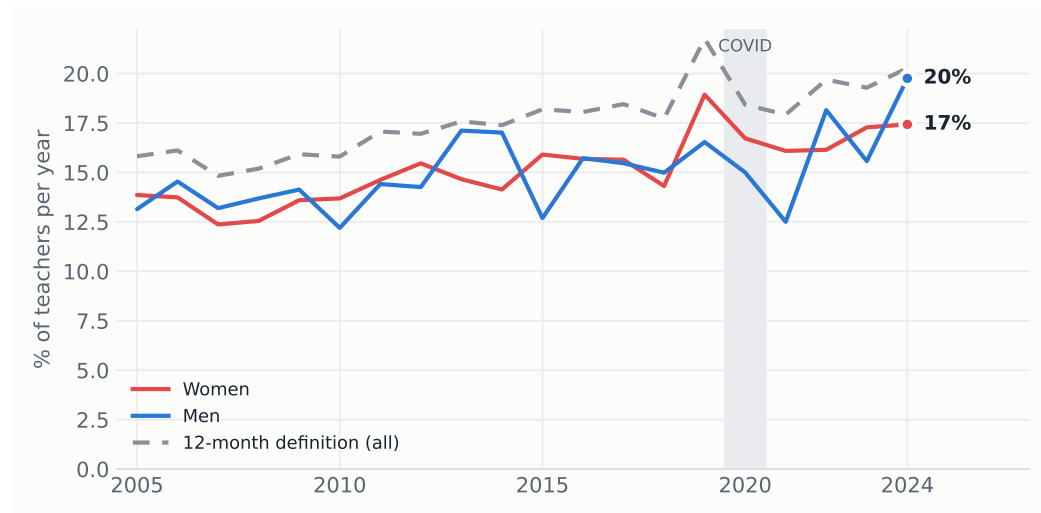
Figure 7: States with better NAEP scores lose fewer teachers.



Notes: Each dot is a state, sized by its number of linked teachers. The horizontal axis averages the official NAEP grade-8 mathematics scale score over the 2005–2024 state assessments, retrieved from the Nation's Report Card data service; the vertical axis is the weighted persistent-attrition rate pooled over 2005–2025. The fitted line is weighted by sample size. Source: authors' calculations from linked CPS pairs and NAEP.

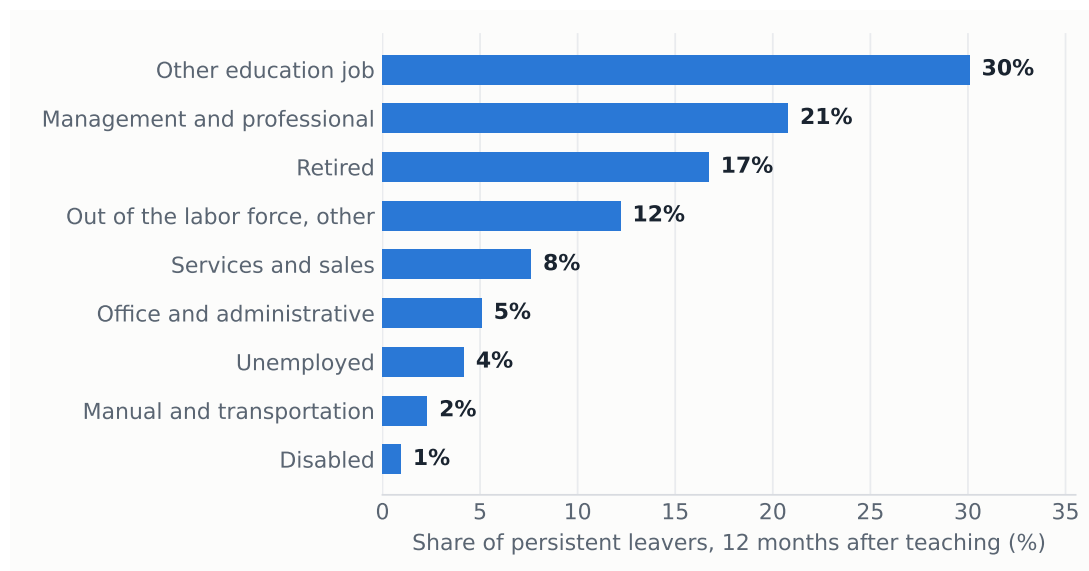
destination, at 30 percent of persistent leavers, is another education job, whether postsecondary teaching, tutoring and instruction, teaching assistance, librarianship or school administration, so a large share of exits from classroom teaching never leave education at all. Retirement accounts for a further 17 percent, at a mean age of 62, and only 4 percent are unemployed a year later. The gender contrast is sharp exactly where the family results of the model will point, since 14 percent of women who leave are out of the labor force for reasons other than retirement or disability, at a mean age of 39 consistent with family care, against 5 percent of men, while men move more often into management and professional jobs, 26 against 19 percent.

Figure 8: Persistent attrition by year and gender.



Notes: The return window is capped by the survey design at three monthly interviews after the twelve-month link, and 21,873 of the 30,244 leavers are re-interviewed at least once. Source: authors' calculations from linked CPS pairs, 2005–2025.

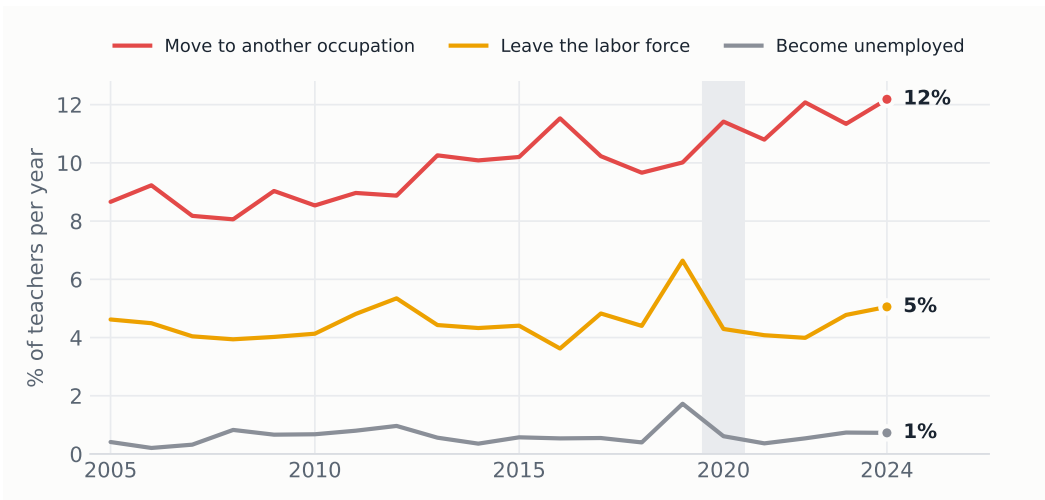
Figure 9: Situation of persistent leavers twelve months after teaching.



Notes: Education jobs other than school teaching include postsecondary teaching, tutoring and instruction, teaching assistance, librarians and education administration, and the remaining employed leavers are grouped into broad occupation classes. Retirement, disability, unemployment and other non-participation come from the labor force status recorded at the second interview. Source: authors' calculations from linked CPS pairs, 2005–2025.

The two margins have also moved differently over time. As Figure 10 makes clear, the rise in attrition documented in Section 3 is almost entirely a rise in occupation switching, from about 8.5 percent of teachers per year in the mid 2000s to 12 percent in 2022–2024, while exits into non-participation have stayed flat at 4 to 5 percent and unemployment spikes only in recessions. Teaching is increasingly losing people to other jobs, not to inactivity.

Figure 10: Leavers by destination over time.

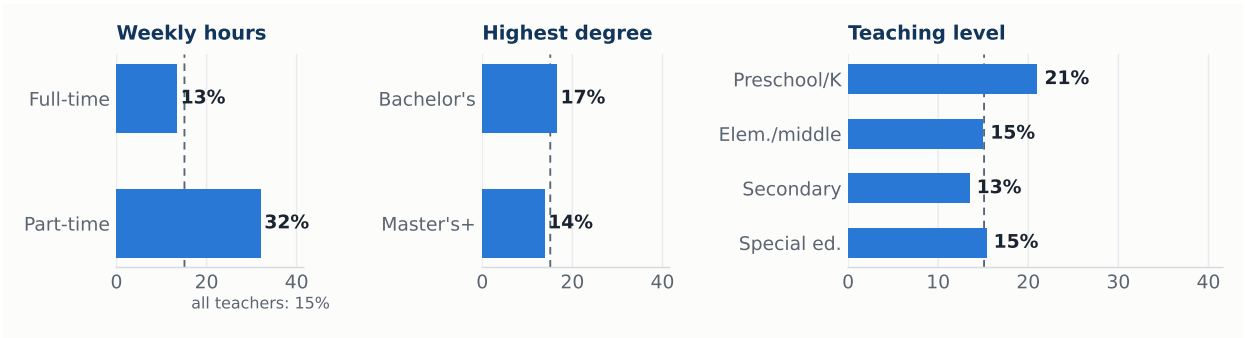


Notes: Persistent leavers as a share of baseline teachers in each year, split by their situation twelve months later.
Source: authors’ calculations from linked CPS pairs, 2005–2025.

5. Who leaves

Raw attrition concentrates along three margins, hours, teaching level and, more weakly among degree holders, credentials. As Figure 11 shows, part-time teachers leave at more than twice the rate of full-time teachers, 32 against 13 percent, and preschool and kindergarten is the most exposed level.

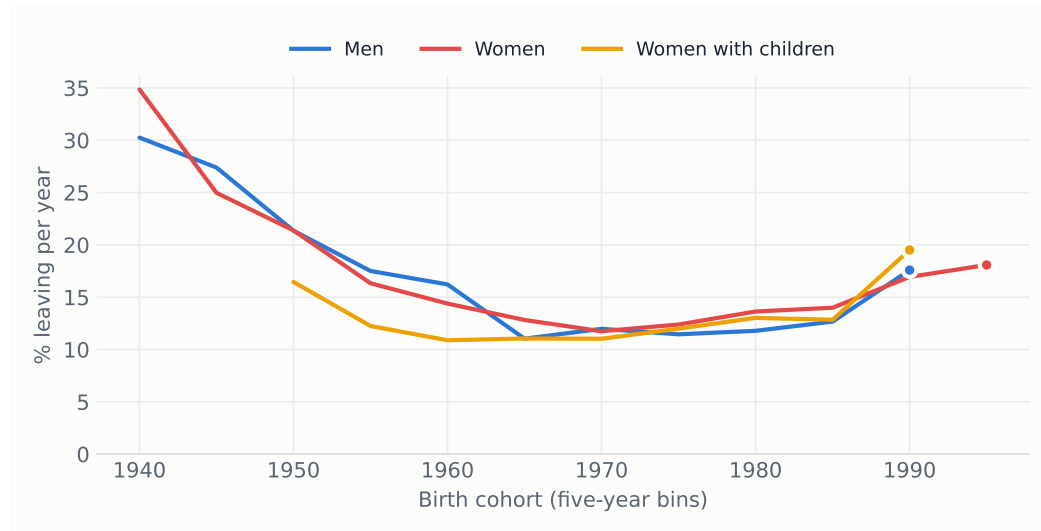
Figure 11: Persistent attrition by baseline characteristics, 2005–2025.



Notes: The dashed line marks the all-teacher average of 15.1 percent. Source: authors’ calculations from linked CPS pairs, 2005–2025.

The lifecycle shape is visible in the raw data as well. Plotting attrition by birth cohort separately for men and women in Figure 12 traces the same U that the model will estimate, with cohorts born before 1950, observed near retirement, leaving at 25 to 35 percent per year, mid-career cohorts at 11 to 13 percent, and the youngest climbing back to 18 percent. Men and women follow nearly the same curve, and the third line explains why the aggregate female curve hides the fertility margin. Among women with children at home, mid-life cohorts are the most stable group in the whole figure, a selection effect, while the youngest cohort of mothers leaves at almost 20 percent, the highest rate shown. Motherhood splits women into the most attached group and the most exit-prone one, and the average conceals both.

Figure 12: Attrition by birth cohort and sex.



Notes: Weighted persistent-attrition rate by five-year birth cohort, pooling all twenty waves, so each cohort is observed at the ages it reaches during 2005–2025 and the curve mixes age and cohort variation. Women with children are those with at least one own child under eighteen at home at baseline, and cells with fewer than 400 observations are omitted. Source: authors' calculations from linked CPS pairs, 2005–2025.

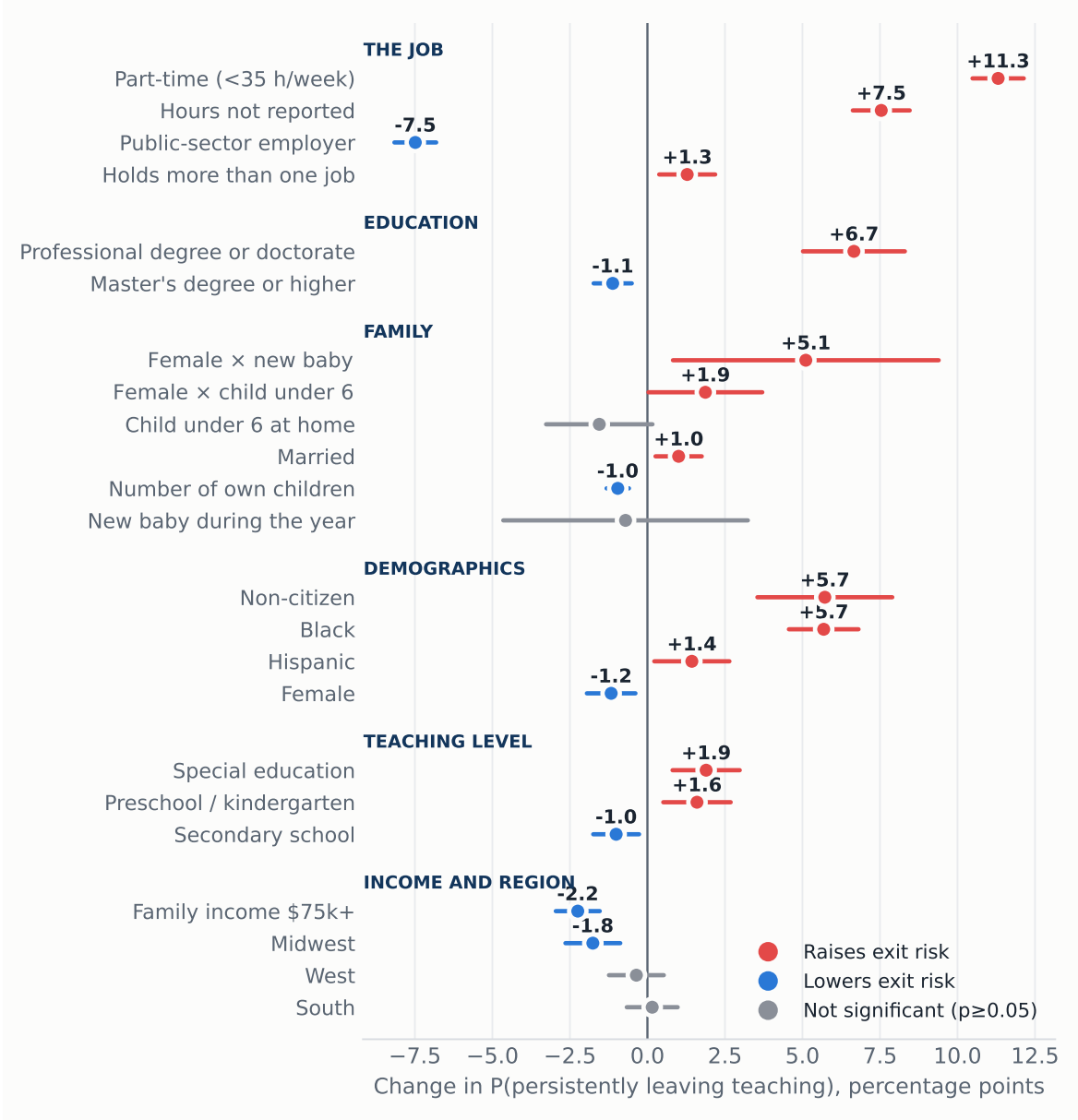
6. The model

We estimate a probit for the probability of persistently leaving, with all covariates measured at baseline, base-year fixed effects and standard errors clustered by household. Table 4 reports the estimates and Figure 13 plots the average marginal effects, organized by the block each variable belongs to.

Five results stand out. First, the job margin dominates, since part-time status raises exit probability by 11.3 points and private-sector employment by 7.5 points relative to public schools, the two largest effects in the model. Second, pay retains. Weekly earnings are only collected in the outgoing rotations, so we estimate the wage effect on that subsample of 44,823 teachers, where ten percent higher weekly earnings lower the probability of leaving by 0.3 points. Third, the family results separate stocks from events. The stock of children retains, at about one point less exit risk per own child at home, while the event of a birth does the opposite and only for women, since a new baby during the year leaves men's exit probability unchanged, a precisely estimated zero, but raises women's by 5.1 points, and 58 percent of the new mothers who leave exit the labor force entirely against 30 percent of the average leaver. This is the fertility channel the literature emphasizes, invisible when only the stock of children is used. Fourth, minority and non-citizen teachers are significantly more likely to leave, with Black teachers at +5.7 points, non-citizens at +5.7 and Hispanics at +1.4, a retention gap that matters for the diversity of the workforce. Fifth, credentials cut both ways, because relative to a bachelor's degree a master's retains modestly while a professional degree or doctorate strongly predicts leaving, at +6.7 points, consistent with better options outside the classroom. Age follows the familiar U, falling to a minimum of 9 percent around age 39 and rising steeply after 55 (the predicted profile is shown in Appendix Figure A2). Every effect is nearly identical under the conventional 12-month definition, with part-time at +13.6 against +11.3, so the predictors do not depend on how leaving is measured.

Estimating the same model on subsamples, as Figure 14 does, confirms that the job effects are strikingly universal, with part-time work raising exit risk by 11 to 13 points for women and men, in

Figure 13: Average marginal effects on the probability of persistently leaving teaching, with 95% confidence intervals.



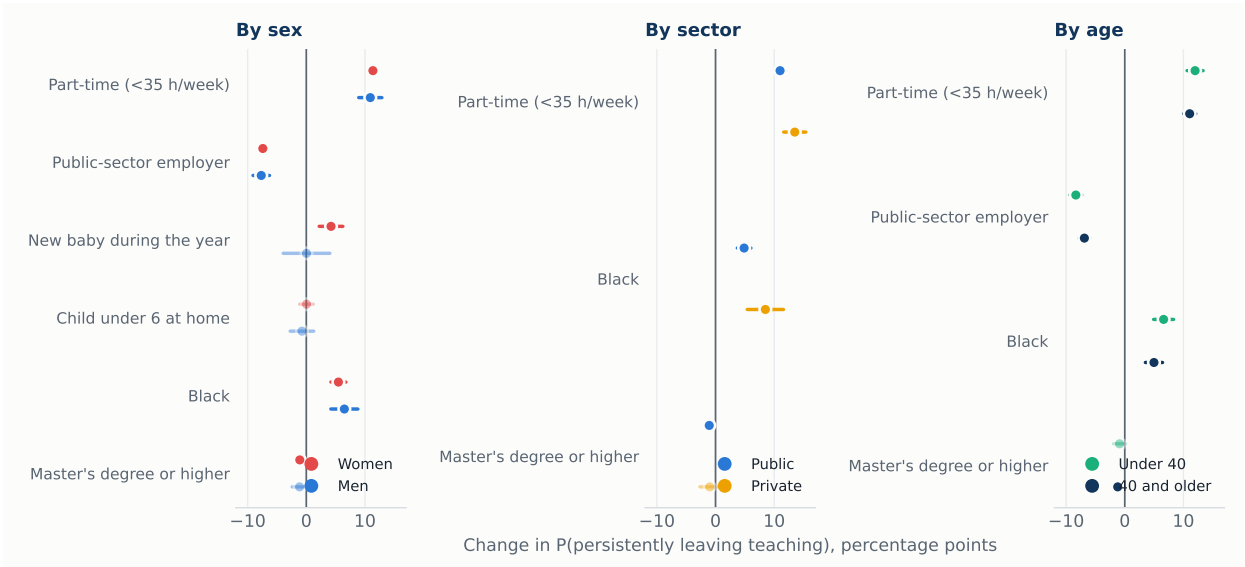
Notes: Estimates from the probit of Table 4. The age profile is shown in Appendix Figure A2.

public and private schools, below and above age 40, and the public-sector advantage at 7 to 8 points in every cut. The fertility effect, in contrast, is entirely female, and the Black retention gap appears in every subsample, largest in private schools.

7. The typical leaver

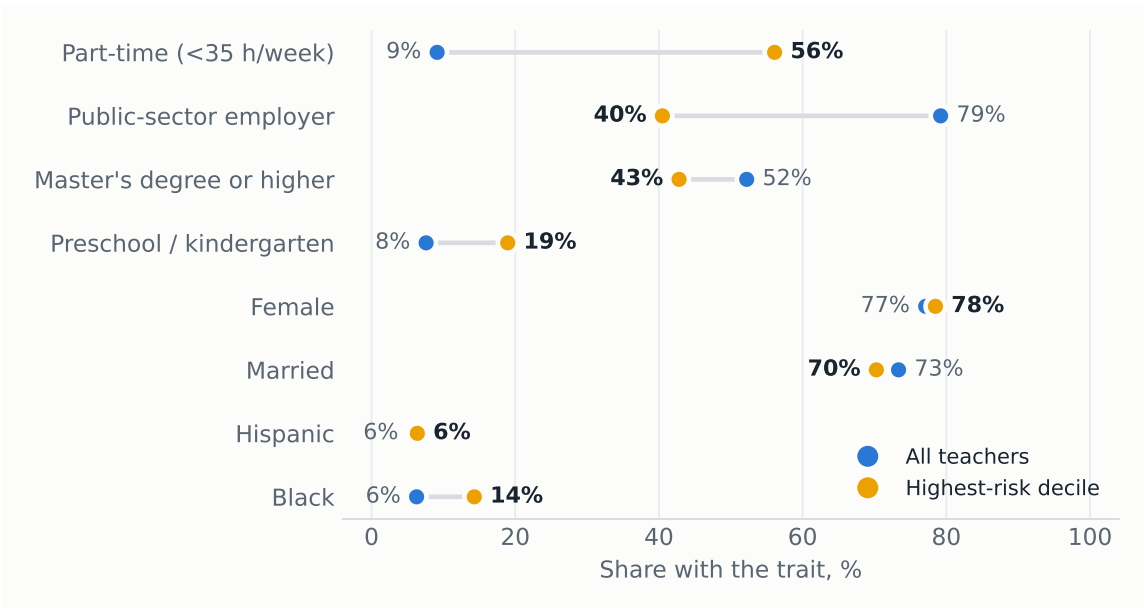
Ranking teachers by their predicted risk gives the model’s portrait of the typical leaver, drawn in Figure 15. The top decile faces a mean exit probability of 37 percent, 2.5 times the average teacher, and it is overwhelmingly female, at 79 percent, late-career with a mean age of 55, six times more likely to work part-time, 56 against 9 percent, and much less anchored in the public sector, where only 40 percent of the group works against 78 percent of all teachers, with Black

Figure 14: The same model estimated on subsamples.



Notes: Average marginal effects with 95% confidence intervals from the probit of Table 4 estimated separately on each subsample; lighter marks are not significant at the 5 percent level. The birth and young-child effects are shown only in the sex panel because they are sex-specific. Source: authors' calculations from linked CPS pairs, 2005–2025.

Figure 15: Characteristics of the top predicted-risk decile vs all teachers.



Notes: Predicted probabilities from the probit of Table 4. Source: authors' calculations from linked CPS pairs, 2005–2025.

teachers over-represented at 14 against 6 percent.

8. Policy implications

The results locate attrition in weak attachment to the job, in part-time hours, private-sector positions and both career extremes, rather than in broad demographics, and they show a structurally rising trend that consists almost entirely of teachers moving to other occupations. Four levers follow directly. Converting involuntary part-time positions to full-time attacks the single largest risk

Table 4: Probit estimates: probability of persistently leaving teaching, 2005–2025.

	Probit coef.	(SE)	AME (pp)
Age (years)	-0.053***	(0.004)	-1.14***
Age ² /100	0.068***	(0.004)	+1.46***
Female	-0.055***	(0.019)	-1.17***
Married	0.047***	(0.018)	+1.01***
Number of own children	-0.045***	(0.009)	-0.96***
Child under 6 at home	-0.072*	(0.041)	-1.55*
Female × child under 6	0.087**	(0.044)	+1.87**
New baby during the year	-0.033	(0.094)	-0.71
Female × new baby	0.238**	(0.102)	+5.11**
Black	0.265***	(0.027)	+5.68***
Hispanic	0.067**	(0.029)	+1.43**
Non-citizen	0.267***	(0.052)	+5.72***
Master's degree or higher	-0.052***	(0.015)	-1.12***
Professional degree or doctorate	0.311***	(0.039)	+6.66***
Part-time (<35 h/week)	0.528***	(0.020)	+11.31***
Hours not reported	0.352***	(0.022)	+7.54***
Holds more than one job	0.060***	(0.021)	+1.28***
Public-sector employer	-0.349***	(0.016)	-7.49***
Family income \$75k+	-0.105***	(0.017)	-2.25***
Midwest	-0.082***	(0.021)	-1.76***
South	0.007	(0.020)	+0.15
West	-0.017	(0.021)	-0.36
Preschool / kindergarten	0.075***	(0.026)	+1.60***
Secondary school	-0.047***	(0.018)	-1.01***
Special education	0.088***	(0.026)	+1.89***
Constant	0.094	(0.096)	
Base-year fixed effects		Yes	
Observations		129,897	
Pseudo R^2		0.068	

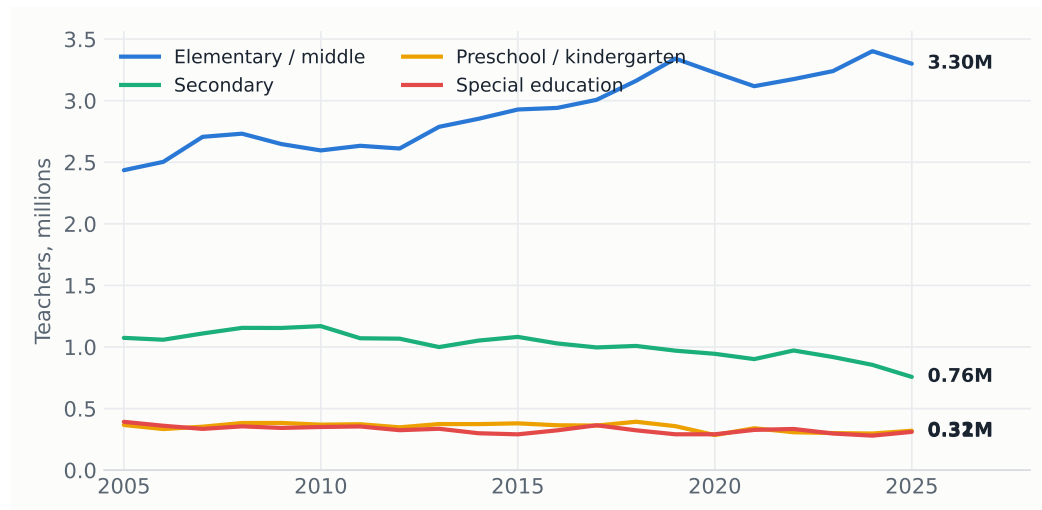
Notes: The outcome is the persistent leaver defined in Section 1. Standard errors, in parentheses, are clustered by household. The AME column reports average marginal effects in percentage points. The sample is all school teachers with a bachelor's degree or higher whose rotation includes at least one interview after the twelve-month link. Reference categories are full-time hours, a bachelor's degree, a private-sector employer, elementary or middle school, and the Northeast region. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

factor, and the wage result implies that pay raises retain, though modestly. The private sector is where teaching careers break, since private-school teachers leave at far higher rates and part of the preschool exposure reflects that composition. The Black, Hispanic and non-citizen retention gaps deserve specific monitoring because they compound into the diversity of the future workforce. And with one in six leavers back within three months, cheap re-entry pathways such as fast-track rehiring, credential portability and return bonuses can recover part of the outflow at low cost.

Methods. Twenty linked CPS 4-8-4 rotation panels, 2005–2025, with 174,872 teachers of whom 129,897 have a follow-up interview. The outcome is the persistent leaver, estimated by probit with base-year fixed effects and household-clustered standard errors. Fully reproducible from the repository scripts.

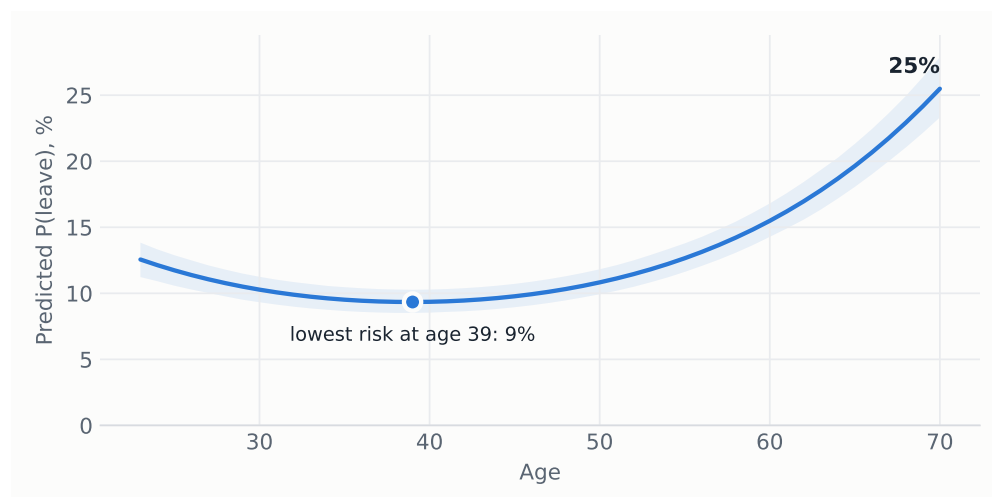
Appendix

Figure A1: Teachers by teaching level.



Notes: Same identification as Figure 4, split by occupation code. Source: authors' calculations from CPS basic monthly microdata, 2005–2025.

Figure A2: Predicted probability of leaving by age.



Notes: Predicted probabilities from the probit of Table 4, holding all other covariates at their sample means, with a simulation-based 95% confidence band.